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DSS/023 – Risk Assessment of Overhead Lines

1 Purpose

The purpose of this document is to detail the company's policy for the risk assessment of overhead lines at all voltage levels (LV up to 132kV) on the UK Distribution networks.

This policy also helps to attain the distribution business goal of:

- Being a leading safety performer within our industry sector

All areas of the business should follow the content of this document when undertaking risk assessment activities associated with overhead lines on the UK distribution networks.

This policy shall be used as the primary source of information for the risk assessment of overhead lines. Guidance on elements of the risk assessment of overhead lines not covered within this policy shall be sought from other appropriate YEDL / NEDL documentation. Section 3.6 lists YEDL / NEDL documents that have been superseded by this and other UKD material.

2 Scope

This document covers the methods and procedures for the risk assessment of overhead lines at all voltages up to and including 132kV.

The document provides guidance on the processes that are to be followed in assigning risk classifications to overhead lines, undertaking risk assessments for each classification, and the minimum mitigation measures to be instigated. This document does not detail specific measures to mitigate risks for all overhead lines but refers to such issues in general terms. Specific risk mitigation solutions shall be developed on an individual basis.

This policy document is aimed at setting policy and offering sources of information to the whole business. It enables all of the different functions within the business to make informed operational decisions that will mitigate the risks associated with overhead lines.

It is envisaged that a company policy for the risk assessment of overhead lines will always be required by the Distribution business. Hence, the drafting intention is to produce a policy that is capable of standing within the company's policy portfolio for an indefinite period.

3 Policy

3.1 Assessment of Relevant Drivers

MidAmerican Energy Holdings Company has made a commitment within its corporate goal statements to be a leading safety performer within our industry sector. In addition, this policy will make a positive contribution to the following business values:

- Safety
- Regulatory, Political and Legal

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3.1.1 Electricity Safety, Quality and Continuity, ESQC, Regulations (2002)

The ESQC Regulations (2002) place obligations on distributors to undertake risk assessments for all overhead lines. They also require that distributors ensure that the public are aware of the dangers associated with overhead lines and that appropriate clearances between overhead lines and other objects are maintained. The subsequent subsections detail the specific regulations relevant to this document.

3.1.1.1 Regulation 3(2)

ESQC Regulation 3(2) requires that;

Generators and distributors shall -

- for each of their overhead lines or part thereof and for each of their substations, assess the foreseeable risk of danger from interference, vandalism or unauthorised access, having regard to both the nature of the equipment and use of the surrounding land, and classify the degree of the risk;
- enter details of the result of the classification of risk in a register or other permanent record kept updated for the purpose; and
- take measures to safeguard the equipment commensurate with the nature and class of risk to which it gives rise.

The DTI guidance notes accompanying the regulations provide substantial information in this area as it is a new regulation. In identifying the risks for overhead lines separate risk assessments are not required for individual supports, equipment attached to supports and the spans between the supports. That said, separate assessments are required for sections of an overhead line where the level of risk may change. In order to satisfy this requirement we will undertake a risk assessment for each support and the associated spans at the intervals detailed in the UKD policy document *MNT/004 - Policy for the Inspection and Maintenance of Overhead Lines*. This is because the equipment on individual supports affects the results of the risk assessment.

In respect of the three areas of risk detailed in paragraph a of this regulation UK Distribution shall consider the following issues:

- Interference – Any accidental contact with overhead line conductors.
- Vandalism – Any wilful act of vandalism which causes damage of any sort to an overhead line.
- Unauthorised access – Any person, other than a competent person, gaining access to the upper levels of a support.

Historically overhead lines have been classed as high risk for reasons other than those above. For example lines crossing motorways and high speed roads were considered high risk due to the consequences associated with the lines falling on to the road and impacting with vehicles. Factors such as these will no longer be referred to as high risk lines. The term high risk shall be reserved for use in respect of the risk of danger from interference, vandalism or unauthorised access.

In relation to assessing the risks the guidance notes indicate that two test should be applied. The first should look at the nature and situation of the equipment. This should assess whether the equipment gives rise to greater risk of interference. For example bare overhead line conductors should be classed as a higher risk than insulated overhead line conductors. The second test should focus on the nature and situation of the surrounding land. This should look at a number of different factors such as evidence of vandalism in the locality, use of mechanised and portable equipment, proximity of features and facilities likely to attract members of the public, and the density and make up of the local population. The overall risk classification should be determined from the higher of the outcome of each of the two tests. This should result in a single measure of risk with at least three levels of risk values.

The guidance notes provide further information on the information to be recorded and the measures which should be taken to mitigate the risks. The measures should be dependent on the risk assessment undertaken and the particular circumstances encountered at each site. The steps could include additional warning signs, enhanced anti-climbing

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guards, or replacement of the equipment with a safer alternative technology, e.g. insulated conductors. The notes also indicate that the risk classification should not be altered due to the mitigating measures put in place, as the potential for danger from interference will remain. This is due to the fact that the mitigating measures may not actually prevent interference in the circumstances and may themselves be subject to vandalism or be removed.

This policy document details our approach to this regulation in respect of Overhead lines.

3.1.1.2 Regulation 3(3)

ESQC Regulation 3(3) requires that;

‘Generators and distributors shall take reasonable steps to ensure that the public are made aware of dangers which may arise from activities carried out in proximity to overhead lines, and to indicate the means by which those dangers may be avoided.’

The guidance notes indicate that the regulation has two requirements. There is a general application requiring communication of the dangers to the public through newsletters, media pieces and local presentations. There is also a specific application which targets specific lines and the groups at risk from accidental contact with those lines. This requires that appropriate measures are taken to draw the attention of the public to the hazards associated with the equipment and how accidental contact can be avoided.

The measures required to meet the specific application of regulation 3(3) are detailed within this policy document. These measures require the application of additional signs at specific sites. The detailed requirements for the additional signs are detailed within the UKD policy document *MNT/003 – Policy for Labelling of Operational Assets*.

3.1.1.3 Regulation 18(5)

ESQC Regulation 18(5) requires that;

‘No overhead line shall, so far is reasonably practicable, come so close to any building, tree or structure as to cause danger.’

In practice this means that we are required to maintain clearances from objects in line with the current company policy, as detailed in YEDL document¹ *DSS/004/037 - Policy for Overhead Line Clearances*. With respect to this policy the risk assessments undertaken at each site shall include a check that the clearances are in accordance with company policy.

3.2 Key Policy Requirements

The objectives of this policy are:

- To ensure the general safety of the public, including third parties undertaking activities in the vicinity of our overhead lines.
- To prevent UK Distribution from having a major breach of legal compliance through interference, of any kind, with our overhead lines.
- To specify the requirements for all risk assessments of Distribution overhead lines for voltages up to and including 132kV.

¹ It is intended that this document form the basis of a harmonised policy document for UK Distribution. Upon publication of the harmonised document this policy document, DSS/023, will be updated to include the appropriate reference. Until this time the YEDL document shall be used as the reference document.

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3.2.1 Risk Classifications

To comply with the requirements of the ESQC regulations each overhead line needs to be classified in terms of the risk of interference, vandalism or unauthorised access. Different risk classifications must be assigned to different parts of the overhead line if the risks associated with that part of the line vary. In assigning the risk for a section of overhead line it is implicit that the level of risk at each support is assessed, so that the points at which the risk classification changes can be clearly identified.

The DTI guidance document supporting the ESQC regulations recommend that two tests be applied which together may be used to assess the risks at that site. The first tests the nature and situation of the equipment and the second tests the nature and situation of surrounding land. This is because different types of equipment will give rise to different levels of risks, as will different activities in the locality of the site.

The guidance document is also strongly persuasive in terms of the level and number of classifications. It recommends that there is a single measure of overall risk and that at least three levels of risk are operated. It goes on to state that the overall risk classification should be determined from the higher of the outcomes of each of the two tests.

With the guidance notes in mind we have chosen to operate three levels of risk, high, medium, and low. Following the 'equipment' and 'surrounding land' tests a risk level of high, medium, or low will be allocated for each test. The higher score from these two tests will be used to provide the overall risk classification. For example a high score in one combined with a low score in the other shall provide an overall level of high risk.

3.2.1.1 Classification for Equipment

The type of equipment installed on overhead lines varies according to the voltage at which the line is operating. In assessing the equipment it is important to remember that we wish to classify overhead lines with respect to the risk of interference from a member of the public (either inadvertent or deliberate). This leads to consideration of how the public could come into contact with the overhead conductors, through climbing the supports.

To classify equipment in this manner we have considered the level of difficulty in climbing a support and how 'desirable' a support would be. Therefore the risk classification is directly related to attributes of the support which vary on a support by support basis. To ensure we capture these variations a risk assessment of individual supports is required.

The table below provides the level of risk classification for each voltage level and type of equipment. The assessment has included all aspects of the overhead line design; conductors, fittings, supports, stays, anti-climbing devices and support mounted equipment. Each overhead line support shall be classified according to the categories outlined in the table below.

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		Line Construction				
		LV ABC all support types	LV Open Wire all support types	Wood Pole Lines HV and EHV	Woodhouse Girder Mast all voltages	Steel lattice mast/ tower all voltages
Support Arrangement	'A' and 'H' Type Supports	Low	Medium	Medium	Medium	High
	Stayed Supports	Low	Medium	Medium	Medium	High
	Supports with Cable Terminations	Low	Medium	Medium	Medium	High
	Supports with Pole Mounted Equipment (Tx's, ABSD, fuses, cabinets etc)	Low	Medium	Medium	Medium	High
	All other supports	Low	Low	Low	Low	High

The definition of all other supports includes wall boxes and wall brackets utilised for terminating services on to buildings. All surface wiring shall be classified as medium risk unless the wiring is ABC, in which case it shall be classified as low risk. The scope of surface wiring includes surface wiring which is fed from underground cable terminations.

3.2.1.2 Classification for Surrounding Land

In contrast to the previous section the use of the surrounding land does not vary according to voltage. Therefore the same test shall be applied at all voltage levels, LV, HV and EHV. In considering the risk from the surrounding land it is important to remember what risk we are assessing:

- Risk of danger from interference, vandalism or unauthorised access

The following three categories have been identified as the general areas which could make an overhead line prone to interference (either inadvertent or deliberate) from the public.

- Recreational Areas
- Vocational Areas
- Local Environment Issues

The following subsections describe the risk classifications for each of these areas. Any span of overhead line falling into one of these categories is to be classified as high or medium according to the details within that subsection. Where a span falls into more than one of these three categories then that span shall be classified as a high 'surrounding land' risk. Where a span does not fall into any of these three categories then that span shall be classified as a low 'surrounding land' risk.

In each category the term "close proximity" relates to an overhead line that runs along side, within 5m of, the area under discussion and could potentially interfere with that area, e.g. if it were to collapse.

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3.2.1.2.1 Recreational Areas

Recreational sites are areas where members of the public regularly congregate to undertake leisure activities. Due to the proximity to the public these overhead lines are at risk of both inadvertent and deliberate interference. Any span of overhead line falling within this category shall be classified according to the table below.

Type of Recreational Area	Overhead Line Crosses Area	Overhead Line in Close Proximity to Area
Children's playgrounds	High Risk	High Risk
Fishing and Sailing locations (inc. Windsurfing)	High Risk	High Risk
Camping and Caravan Sites	High Risk	High Risk
Common land used for recreational activities (e.g. parkland, kite flying, wasteland where youths gather etc)	High Risk	Medium Risk
Allotments	Medium Risk	Medium Risk
Sports grounds and stadiums	Medium Risk	Medium Risk
Golf Courses	Medium Risk	Medium Risk

3.2.1.2.2 Vocational Areas

Vocational areas are where some members of the public are likely to undertake work activities within the vicinity of an overhead line. Due to the proximity to the public these overhead lines are at risk of both inadvertent and deliberate interference. Any span of overhead line falling within this category shall be classified according to the table below.

Type of Vocational Area	Overhead Line Crosses Area	Overhead Line in Close Proximity to Area
Schools	High Risk	High Risk
Business premises with portable machinery	High Risk	Medium Risk
Arable land where farm machinery is used	Medium Risk	Medium Risk

For the business premises with portable machinery grouping we are concerned with inadvertent contact with the overhead line from the machinery used at the site. Examples of this would be farmyards, sites where vehicles are loading using fork lift trucks, sites where tipper truckers are operated, quarries, sites where cranes are used. These examples are by no means exhaustive, it merely provides an indication of the sites covered by this grouping.

3.2.1.2.3 Local Environment Issues

In some areas there are local environment issues which create an environment where some members of the public are likely to congregate within the vicinity of an overhead line. Due to the proximity to the public these overhead lines are at risk of both inadvertent and deliberate interference. Any span or spans of overhead line falling within this category shall be classified according to the table below.

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Type of Local Environment	Overhead Line Crosses Area	Overhead Line in Close Proximity to Area
Areas with potential climbing aids	High Risk	High Risk
“Travellers” sites	High Risk	High Risk
Areas prone to vandalism	High Risk	High Risk
Domestic Private Gardens	High Risk	Medium Risk
Footpaths (inc. Disused Railways)	Medium Risk	Medium Risk

The proximity of vegetation to the overhead line was also considered as a local environment factor affecting the level of risk. Where vegetation with climbable limbs encroaches within the statutory clearance from overhead lines the potential for a person to come into contact with the overhead line is increased. The UKD policy document *DSS/024 - Policy for the Management and Control of Vegetation near Overhead Lines* details the policy for mitigation of the risks associated with vegetation encroachment. In particular the document details the maximum permissible timescales for the removal of encroaching vegetation. Shorter timescales are prescribed for high risk sites due to the increased risk with members of the public in the locality. Therefore vegetation encroachment has not been included as a factor contributing to the level of risk associated with an overhead line.

3.2.1.3 Overall Classifications

The overall risk assessment is based on the classifications for the nature and situation of the surrounding land and the nature and situation of equipment. The table below demonstrates how the classifications will be combined to provide an overall level of risk for the relevant section of overhead line. Within the overall assessment equal weightings are given to the two tests.

Although the higher of the two tests has been used to set the overall risk classifications a high risk from one assessment does not in itself result in an overall risk classification of high. For example a high risk for the ‘equipment’ test combined with a low risk for ‘surrounding land’ test would result in an overall risk classification of medium.

Nature and Situation of Surrounding Land Classification	Nature and Situation of Equipment Classification			
		High	Medium	Low
	High	High	High	High
	Medium	High	Medium	Medium
	Low	High	Medium	Low

On completion of the overall risk assessment, all overhead lines with an overall rating of high or medium should have the mitigation measures implemented as outlined in section 3.2.2 of this document.

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3.2.1.4 Method of Classifications

3.2.1.4.1 Ongoing Classifications

An inspection of each overhead line is needed to assign to the appropriate risk classification. During the inspection a risk assessment of each support shall be undertaken. In each case this will require a review of the equipment on the section of overhead line and the usage of the surrounding land. The classifications shall be undertaken at the minimum inspection interval which covers all assets of a particular type. The inspections shall be undertaken at the intervals detailed in the UKD policy document *MNT/004 - Policy for the Inspection and Maintenance of Overhead Lines*. The classifications are included as a standard part of the appropriate inspection.

For all HV and EHV overhead lines this can be achieved from the helicopter patrol. During the helicopter patrol each overhead line, or section of line as appropriate, will be classified using the checklist provided in appendix A. In some cases a full classification may not be achievable from the helicopter. In these cases a foot patrol of that section of line shall be undertaken to ensure that all overhead lines are classified appropriately.

LV overhead lines are not inspected through a helicopter patrol due to their location and proximity to the public. Each LV overhead line, or section of line, is inspected through a ground patrol. In general there are two types of LV overhead line; LV open wire lines and LV ABC lines. Currently there are different procedures, and therefore minimum intervals, covering these two asset groups. The inspection with the minimum interval covering all LV open wire lines is the proximity inspection. For LV ABC lines the inspection with the minimum interval covering all lines is the ground patrol. All LV lines will be classified during the inspections detailed above using the checklist provided in appendix B.

The table below details the method of classification for each asset group and the checklist which shall be used.

Activity	Asset Group			
	LV Open Wire	LV ABC	HV	EHV
Helicopter Patrol			Checklist A	Checklist A
Proximity Inspection	Checklist B			
Ground Patrol		Checklist B		

3.2.1.4.2 Initial Classifications

The ESQC Regulations require that risk classifications are completed on all overhead lines by 31st January 2008. These timescales would not be achieved for LV overhead lines using the methods outlined in the previous section. Therefore another method shall be used for the initial risk classification of LV overhead lines.

During 2004 a survey of the LV overhead network is being undertaken to assess the proximity of vegetation. The scope of the survey shall include the initial risk classification of LV overhead lines.

3.2.2 Standard Mitigating Measures

At each high and medium risk site mitigating measures shall be implemented to make the public aware of the dangers and to mitigate the risk of interference whether accidental or deliberate. The table in appendix E details the standard mitigating measures to be applied at each type of high and medium risk site.

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3.2.2.1 Application of Additional Safety Signs

Additional safety signs shall be fitted to all supports at all high and medium risk sites in accordance with sections 3.2.4.1.3 and 3.2.4.2.2 of the UKD policy document *MNT/003 – Policy for Labelling of Operational Assets*. Sufficient signs shall be fitted to form a continuous ring around the support. These signs are required to make the public accessing the area aware of the dangers associated with activities in the vicinity of overhead lines.

3.2.2.2 Inspection and Risk Assessment of Sites

All high risk sites shall be inspected at the intervals detailed in the UKD policy document *MNT/004 - Policy for the Inspection and Maintenance of Overhead Lines*. These inspections are undertaken to ensure that the risk associated with the site is re-assessed and to ensure that the mitigating measures, which have been implemented, remain sufficient to deter unauthorised access. The specific requirements for these inspections are detailed in section 3.2.3.2 of this document.

3.2.2.3 Application of Anti-Climbing Devices

Enhanced anti climbing guards shall be fitted to all supports within recreational and local environment high risk sites. They shall also be fitted to all supports designated as high risk due to their proximity to schools. These will minimise the risk of a member of the public gaining unauthorised access to the upper levels of the support. Where enhanced anti-climbing guards are applied to lattice towers they shall be fitted on each tower leg.

Anti climbing barbed wire shall be fitted to all stays within the sites referred to above. These will minimise the risk of a member of the public gaining unauthorised access to the upper levels of the support via the stay.

Cable guards shall be fitted to all cables attached to supports within the site. These will prevent the cable being used as a climbing aid to gain access to the upper levels of the supports.

3.2.2.4 Inadvertent Contact Signs

At some locations the risk of inadvertent contact with the overhead line is increased due to the activities undertaken in proximity to the site. For example at fishing locations there is increased risk of contact from fishing lines during casting. At these sites inadvertent contact warning signs shall be erected.

Where practicable inadvertent contact warning signs shall be erected at all recreational high and medium risk sites and at all vocational high risk sites. For clarity this covers the following high and medium risk sites;

- Children's playgrounds
- Fishing and Sailing locations
- Camping and Caravan sites
- Common land used for kite flying
- Schools
- Business premises with portable machinery (high risk only)

For the business premises with portable machinery grouping we are concerned with inadvertent contact with the overhead line from the machinery used at the site. Examples of this would be farmyards, sites where vehicles are loading using fork lift trucks, sites where tipper trucks are operated, quarries, sites where cranes are used. These examples are by no means exhaustive, it merely provides an indication of the sites covered by this grouping.

Warning and prohibition signs shall be fitted at the sites detailed above in accordance with sections 3.2.4.1.4 and 3.2.4.2.3 of the UKD policy document *MNT/003 – Policy for Labelling of Operational Assets*. The permission of the site owner will be obtained prior to erection of the signs. If the site owner withholds their permission this information will be recorded against the high risk site in the appropriate system.

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Where inadvertent contact signs are erected they shall be inspected at the intervals detailed in the UKD policy document *MNT/004 - Policy for the Inspection and Maintenance of Overhead Lines*. The specific requirements for these inspections are detailed in section 3.2.3.3 of this document.

3.2.2.5 Further Mitigating Measures

At some locations the mitigating measures detailed in the previous sections may be viewed to be insufficient to manage the assessed level of risk. In these cases consideration must be given to the application of further mitigating measures. The types of measures which should be considered are

- Undertaking more frequent risk inspections at that site
- Installation of enhanced anti-climbing guards
- Instigation of communications with the site owner and site users
- Temporary shrouding of the conductors (specific cases only)
- Increasing height of overhead line
- Use of covered or insulated conductor
- Diversion of the overhead line
- Undergrounding of the overhead line

The use of further mitigating measures shall be assessed on a site by site basis. The selection of appropriate measures shall take into account various factors including the level of risk presented at the site, the degree of mitigation afforded by the proposed measures, the cost of the proposed measures and the practicalities of implementation at that site. Where there is doubt about what measures should be applied the solution should be identified through consultation between the Construction Design, Safety and Network Investment sections.

Temporary shrouding of the line shall only be utilised in cases where a temporary structure results in the infringement of clearances to the overhead line. For example if scaffolding is erected while work is being undertaken on a house and the clearances to the overhead line are infringed. If there is any doubt about the appropriateness of using temporary shrouding the solution shall be identified through consultation between the Construction Design and Safety sections.

3.2.3 Risk Classification and Assessment Procedures

All risk assessments shall be undertaken during a ground patrol of the overhead line. A standard procedure to confirm the risk classification shall be used for all overhead lines. This procedure shall be supplemented by an additional risk assessment procedure for high risk sites. This procedure will be further supplemented by an inadvertent contact assessment for all sites where inadvertent contact signs are required. These procedures shall be undertaken at the intervals prescribed within the UKD policy document *MNT/004 - Policy for the Inspection and Maintenance of Overhead Lines*.

3.2.3.1 Risk Classification Procedure

The purpose of this procedure is to confirm the risk classification of the overhead line, or section of overhead line. The procedure shall only be used for the classification of LV overhead lines from a foot patrol. The procedure addresses the classification of the equipment and the classification of the surrounding land. The following questions shall be asked. The data collection sheet for this procedure is detailed in appendix B. The standard responses to be used in each section of the procedure are detailed at the start of that section.

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3.2.3.1.1 Equipment Classification

For each of the following questions a standard response of *Yes* or *No* is required. A *No* response will indicate that the support has some equipment which will contribute to the overall risk classification.

- Is the support of a construction other than 'A' or 'H' type?
- Is the support free of stays attached to it?
- Is the support free of cable terminations?
- Is the support free of equipment mounted on it which results in less than 3m of 'clean' support above ground level?

3.2.3.1.2 Surrounding Land Classification

For each of the following questions a standard response of *Crosses area*, *In close proximity to area*, or *Not in the immediate area* is required. A *Not in the immediate area* response will indicate that the land surrounding the support does not have an issue which would contribute to the overall risk classification.

In each of these questions the term close proximity relates to an overhead line that runs along side the area under discussion and could potentially interfere with that area, e.g. if it were to collapse.

Recreational Area

- Is this support, or the associated spans, in the vicinity of a children's playground?
- Is this support, or the associated spans, in the vicinity of a fishing or sailing location?
- Is this support, or the associated spans, in the vicinity of a camping or caravan site?
- Is this support, or the associated spans, in the vicinity of common land used for recreational activities?
- Is this support, or the associated spans, in the vicinity of an allotment?
- Is this support, or the associated spans, in the vicinity of sports ground or stadium?
- Is this support, or the associated spans, in the vicinity of a golf course?

Vocational Area

- Is this support, or the associated spans, in the vicinity of a school?
- Is this support, or the associated spans, in the vicinity of business premises with portable machinery?
- Is this support, or the associated spans, in the vicinity of arable land where farm machinery is used?

Local Environment

- Is this support, or the associated spans, in the vicinity of a potential climbing aid?
- Is this support, or the associated spans, in the vicinity of a 'travellers' site?
- Is this support, or the associated spans, in the vicinity of an area prone to vandalism?
- Is this support, or the associated spans, in the vicinity of a domestic private garden?
- Is this support, or the associated spans, in the vicinity of a footpath (inc. Disused railways)?

Evidence of vandalism or unauthorised access

For each of the following questions a standard response of *Yes* or *No* is required. A *No* response will indicate that the support has some equipment which will contribute to the overall risk classification.

- Is our equipment free from signs of vandalism, unauthorised access or interference?
- Is the area in the immediate vicinity of the line free of evidence of loitering or gatherings of youths? *Examples might be drink cans, food wrappings, graffiti, cigarette ends etc. Blown debris should be discounted.*
- Is there any equipment or materials on site which may be used to provide unauthorised access to the overhead equipment?

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3.2.3.2 Risk Assessment Procedure

The purpose of this procedure is to review the condition of the equipment for any safety issues and to ensure that sufficient mitigating measures are in place to minimise the risk of unauthorised access. This procedure shall be used for the risk assessment of all high risk sites and for the classification of HV and EHV overhead lines for foot patrols in lieu of a helicopter patrol. The questions detailed under the risk classification procedure shall be asked along with the following questions. The questions are worded such that a yes answer indicates there are no issues. The data collection sheet for this procedure is detailed in appendix C.

3.2.3.2.1 Condition of Equipment

- Are the clearances² to buildings, including other objects which can be stood upon, in line with company policy within the site?
Minimum clearance³ should be 3m for open wire conductor under 33kV, 3.2m for 33/66kV, 3.6m for 132kV and 1m for free swinging ABC.
- Are the clearances to ground in line with company policy within the site?
Under 33kV 5.8m to roads and 5.2m in other cases, 33/66kV 6m in all cases, 132kV 6.7m in all cases.
- Are the clearances to trees in line with company policy within the site?
Report if trees infringe within 0.8m under 33kV, 1.0m 33/66kV and 1.4m for 132kV.
- Is the conductor insulation in good condition within the site?
- Is the earthing in good condition within the site?
- Are the supports within the site in good condition?
- Are the stays within the site in good condition?
- Do the stays within the site have stay insulators fitted and located 3m above ground level and below the lowest line on that support?
- Are property labels fitted and in good condition on all supports with plant within the site?
- Do all the supports within the site have the support number plate present and in good condition?

3.2.3.2.2 Assessment of Mitigating Measures

- Are sufficient danger of death signs fitted to all supports within the site?
- Are anti climbing guards fitted, and in good condition, on all supports within the site?
- Is anti climbing barbed wire fitted and in good condition on all stays within the site?
- Are cable guards fitted and in good condition within the site?

3.2.3.3 Inadvertent Contact Assessment Procedure

The purpose of this procedure is to review the mitigating measures in place to minimise the risk of inadvertent contact with the overhead line. The procedure shall only be used for the assessment of sites where inadvertent contact signs should be erected. The questions detailed under the risk classification and risk assessment procedures shall be asked along with the following questions. The questions are worded such that a yes answer indicates there are no issues. The data collection sheet for this procedure is detailed in appendix D.

² Clearances shall be measured using 'rangefinder' or similar equipment. Fuse rods shall not be used, as they are not a suitable measurement device.

³ Clearances to buildings do not apply to surface wiring, or service spans to terminations on building walls.

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3.2.3.3.1 Assessment of Mitigating Measures

- Does the site require inadvertent contact warning signs?
- Does the site have one or more access points?
- Are there warning notices in good condition at each access point?
Signs should be fitted where site conditions permit and agreement of landowner has been obtained
- Are there prohibition signs in good condition at appropriate points within the site?
Where practicable signs should be fitted 30m either side of the line and 50m apart along the line, where site conditions permit and agreement of landowner has been obtained
- Is the site owner present?
- Are the site owner's contact details correct?
- Does the site owner have our emergency contact number?
- Has the site owner refused further advice on safety near electrical lines and equipment?

3.3 Assumptions

This policy is aimed at ensuring that all overhead lines are classified in terms of the foreseeable risk of danger from interference, vandalism or unauthorised access, and that mitigating measures are implemented commensurate with the assessed level of risk.

The following assumptions have been made and will be required to ensure the effective delivery of this policy into the business:

- The results of the risk assessments and the corresponding mitigating measures are stored in the relevant company data systems.
- The additional safety, warning and prohibition signs are appropriate and feasible to be erected at the designated high and medium risk sites.
- The UKD policy for overhead line clearances complies with EATS 43-8 Overhead Line Clearances.

3.4 Implementation and Monitoring of this Policy

Several departments within UK Distribution are accountable for the implementation and monitoring of the risk assessment of overhead lines policy in accordance with the following table:

Designation	Role
Maintenance Manager	Accountable for the implementation of all aspects of the risk assessment of overhead lines policy relevant to maintenance and will ensure responsible persons are appointed to implement this policy.
Construction Manager	Accountable for the implementation of all aspects of the risk assessment of overhead lines policy relevant to construction and will ensure responsible persons are appointed to implement this policy.
Safety Manager	Monitor compliance of those accountable for implementation of the risk assessment of overhead lines policy.

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3.5 Planned Policy Review

This policy shall be proposed for review on a biennial basis or at any time when external or internal influences drive a change in policy e.g. a change in legislation, learning points from the initial implementation stage.

The following responsibilities shall apply to policy control and review:

Designation	Responsibility
Publication Manager	Responsible for issuing a quarterly report to the Policy Production Manager (or representative) detailing policies scheduled for biennial review within the next 6 months
Policy Production Manager	Responsible for assessing the continued applicability of this company policy and for amending this document and communicating any changes in policy.

3.6 Superseded Documentation

There is no documentation that has been superseded in its entirety by this policy.

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4 References

4.1 External Documentation

	Reference	Title	Version and date
1	Statutory Instrument 2002 No. 2665	The Electricity Safety, Quality and Continuity Regulations 2002	2002
2	DTI document	Guidance on the Electricity Safety, Quality and Continuity Regulations	2002
3	EATS 43-8	Electricity Association Technical Specification 43-8 Overhead Line Clearances	Issue 2: 1988 as amended

4.2 Internal Documentation

	Reference	Title	Version and date
1	MNT/003	Policy for Labelling of Operational Assets	V1 Oct 2003
2	MNT/004	Policy for the Inspection and Maintenance of Overhead Lines	V1 Nov 2003
3	DSS/024	Policy for the Management and Control of Vegetation Near Overhead Lines	V1 Nov 2003
4	YEDL – DSS/004/037	Guidance on Overhead Line Clearances	V1 Nov 1999

5 Definitions

Term	Definition
ABC	Aerial Bundled Conductor
ABSD	Air Brake Switch Disconnecter
DTI	Department of Trade and Industry
EHV	Extra High Voltage - Means a voltage greater than or equal to 33,000V
EHV Tower	EHV Supports constructed from metal including lattice towers and steel masts
EHV Wood	EHV Supports constructed from wood
ESQC Regulations	Electricity Safety, Quality and Continuity Regulations
HGV	Heavy Goods Vehicle
HV	High Voltage - Means a voltage greater than 1000V, but less than 33,000V
LV	Low Voltage - Means a voltage up to and including 1000V
NEDL	Northern Electric Distribution Limited
Support	Structure whose purpose is to keep the overhead conductor at a specified height above ground level.
UKD	UK Distribution, consisting of NEDL and YEDL
YEDL	Yorkshire Electricity Distribution plc

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6 Authority for Issue

1. Policy Author

I sign to confirm that I have completed and checked this policy and I am satisfied with its content

		Sign	Date
Stephen Murray	Network Investment Engineer	Stephen Murray	10/10/03

2. Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this policy and submit it for approval and authorisation.

		Sign	Date
Mark Nicholson	Network Policy Manager	Mark Nicholson	10/10/03

3. Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this policy and submit it for approval and authorisation.

		Sign	Date
Mark Drye	Network Integrity Performance Island Lead	Mark Drye	10/10/03
Mick Hickling	Safety & Environment Performance Island Lead	Mick Hickling	10/10/03

4. DBD Assurance

I sign to confirm that this policy has been assured for issue on to the DBD system

		Sign	Date
Sean Johnson	Safety Engineer	Sean Johnson	10/10/03

5. Approval

Approval is given for the content of this policy

		Sign	Date
Peter McCormick	Utilities Safety, Environment and Training Director	Peter McCormick	10/10/03

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6. Authorisation

Authorisation is granted for publication of this policy

		Sign	Date
Phil Jones	Strategy and Investment Director	Phil Jones	12/10/2003

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Appendix A – Checklist for Classification from Helicopter

This checklist shall only be used for the classification of HV and EHV overhead lines from a helicopter patrol.

Helicopter Patrol Risk Classification Checklist			
Classification Area	Type	Risk Level	
Equipment	‘A’ or ‘H’ Type Supports	Medium	
	Stayed Supports	Medium	
	Supports with Cable Terminations	Medium	
	Supports with Pole Mounted Equipment (Tx’s, ABSD, PM fuses, cabinets etc)	Medium	
	Steel lattice mast/ tower	High	
Classification Area	Type	Crosses Area	Proximity to Area
Recreational Area	Children’s playgrounds	High	
	Fishing and Sailing locations (inc. Windsurfing)	High	
	Camping and Caravan Sites	High	
	Common land used for recreational activities (e.g. parkland, kite flying, wasteland where youths gather etc)	High	Medium
	Allotments	Medium	
	Sports grounds and stadiums	Medium	
	Golf Courses	Medium	
Vocational Area	Schools	High	
	Business premises (excluding office premises)	High	Medium
	Arable land where farm machinery is used	Medium	
Local Environment	Areas with potential climbing aids	High	
	“Travellers” sites	High	
	Areas prone to vandalism	High	
	Domestic Private Gardens	High	Medium
	Footpaths (inc. Disused Railways)	Medium	

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Appendix B –Risk Classification Procedure

This procedure shall only be used for the classification of LV overhead lines from a foot patrol.

Risk Classification Data Collection Sheet					
Line No:		Pole Number:		Grid Ref	
Question					Answer
Equipment Classification					
1. Is the support of a construction other than 'A' or 'H' type?					YES/NO
2. Is the support free of stays attached to it?					YES/NO
3. Is the support free of cable terminations?					YES/NO
4. Is the support free of equipment mounted on it which results in less than 3m of 'clean' support above ground level?					YES/NO
Recreational Area					
5. Is this support, or the associated spans, in the vicinity of a children's playground?					Crosses area/ In close proximity to area/ Not in the immediate area
6. Is this support, or the associated spans, in the vicinity of a fishing or sailing location?					
7. Is this support, or the associated spans, in the vicinity of a camping or caravan site?					
8. Is this support, or the associated spans, in the vicinity of common land used for recreational activities?					
9. Is this support, or the associated spans, in the vicinity of an allotment?					
10. Is this support, or the associated spans, in the vicinity of sports ground or stadium?					
11. Is this support, or the associated spans, in the vicinity of a golf course?					
Vocational Area					
12. Is this support, or the associated spans, in the vicinity of a school?					Crosses area/ In close proximity to area/ Not in the immediate area
13. Is this support, or the associated spans, in the vicinity of business premises with portable machinery?					
14. Is this support, or the associated spans, in the vicinity of arable land where farm machinery is used?					
Local Environment					
15. Is this support, or the associated spans, in the vicinity of a potential climbing aid?					Crosses area/ In close proximity to area/ Not in the immediate area
16. Is this support, or the associated spans, in the vicinity of a 'travellers' site?					
17. Is this support, or the associated spans, in the vicinity of an area prone to vandalism?					
18. Is this support, or the associated spans, in the vicinity of a domestic private garden?					
19. Is this support, or the associated spans, in the vicinity of a footpath (inc. Disused railways)?					
Evidence of vandalism or unauthorised access					
20. Is our equipment free from signs of vandalism, unauthorised access or interference?					YES/NO
21. Is the area in the immediate vicinity of the line free of evidence of loitering or gatherings of youths? Examples might be drink cans, food wrappings, graffiti, cigarette ends etc. Blown debris should be discounted.					YES/NO
22. Is there any equipment or materials on site which may be used to provide unauthorised access to the overhead equipment?					YES/NO

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Appendix C – Risk Assessment Procedure

This procedure shall be used for the risk assessment of all high risk sites and for the classification of HV and EHV overhead lines from foot patrols in lieu of a helicopter patrol.

Overhead Line Risk Assessment Data Collection Sheet				
Line No:		Pole Number:		Grid Ref
Question			Answer	Action
Equipment Classification				
1. Is the support of a construction other than 'A' or 'H' type?			YES/NO	None
2. Is the support free of stays attached to it?			YES/NO	None
3. Is the support free of cable terminations?			YES/NO	None
4. Is the support free of equipment mounted on it which results in less than 3m of 'clean' support above ground level?			YES/NO	None
Recreational Area				
5. Is this support, or the associated spans, in the vicinity of a children's playground?			Crosses area/ In close proximity to area/ Not in the immediate area	None
6. Is this support, or the associated spans, in the vicinity of a fishing or sailing location?				
7. Is this support, or the associated spans, in the vicinity of a camping or caravan site?				
8. Is this support, or the associated spans, in the vicinity of common land used for recreational activities?				
9. Is this support, or the associated spans, in the vicinity of an allotment?				
10. Is this support, or the associated spans, in the vicinity of sports ground or stadium?				
11. Is this support, or the associated spans, in the vicinity of a golf course?				
Vocational Area				
12. Is this support, or the associated spans, in the vicinity of a school?			Crosses area/ In close proximity to area/ Not in the immediate area	None
13. Is this support, or the associated spans, in the vicinity of business premises with portable machinery?				
14. Is this support, or the associated spans, in the vicinity of arable land where farm machinery is used?				
Local Environment				
15. Is this support, or the associated spans, in the vicinity of a potential climbing aid?			Crosses area/ In close proximity to area/ Not in the immediate area	None
16. Is this support, or the associated spans, in the vicinity of a 'travellers' site?				
17. Is this support, or the associated spans, in the vicinity of an area prone to vandalism?				
18. Is this support, or the associated spans, in the vicinity of a domestic private garden?				

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Question	Answer	Action
19. Is this support, or the associated spans, in the vicinity of a footpath (inc. Disused railways)?		
Evidence of vandalism or unauthorised access		
20. Is our equipment free from signs of vandalism, unauthorised access or interference?	YES/NO	None
21. Is the area in the immediate vicinity of the line free of evidence of loitering or gatherings of youths? <i>Examples might be drink cans, food wrappings, graffiti, cigarette ends etc. Blown debris should be discounted.</i>	YES/NO	None
22. Is there any equipment or materials on site which may be used to provide unauthorised access to the overhead equipment?	YES/NO	None
Condition of Equipment		
23. Are the clearances to buildings in line with company policy within the site?	YES/NO	Rectify if dangerous
24. Are the clearances to ground in line with company policy within the site?	YES/NO	Rectify if dangerous
25. Are the clearances to trees in line with company policy within the site?	YES/NO	Rectify if dangerous
26. Is the conductor insulation in good condition within the site?	YES/NO	Note or repair as appropriate
27. Is the earthing in good condition within the site?	YES/NO/NOT AT POLE	Note or repair as appropriate
28. Are the supports within the site in good condition?	YES/NO	Note or repair as appropriate
29. Are the stays within the site in good condition?	YES/NO/NOT AT POLE	Note or repair as appropriate
30. Do the stays within the site have stay insulators fitted and located 3m above ground level and below the lowest line on that support?	YES/NO/NOT AT POLE	Note or fit as appropriate
31. Do all the supports within the site have the support number plate present and in good condition?	YES/NO/NOT AT POLE	Note or fit as appropriate
Assessment of Mitigating Measures		
32. Are sufficient danger of death signs fitted to all supports within the site?	YES/NO	Note or fit as appropriate
33. Are anti climbing guards fitted, and in good condition, on all supports within the site?	YES/NO	Note or fit as appropriate
34. Is anti climbing barbed wire fitted and in good condition on all stays within the site?	YES/NO	Note or fit as appropriate
35. Are cable guards fitted and in good condition within the site?	YES/NO	Note or fit as appropriate

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Appendix D – Inadvertent Contact Assessment Procedure

This procedure shall only be used for the assessment of sites where inadvertent contact signs should be erected.

Inadvertent Contact Assessment Data Collection Sheet					
Line No:		Pole Number:		Grid Ref	
Question				Answer	Action
Equipment Classification					
1. Is the support of a construction other than 'A' or 'H' type?				YES/NO	None
2. Is the support free of stays attached to it?				YES/NO	None
3. Is the support free of cable terminations?				YES/NO	None
4. Is the support free of equipment mounted on it which results in less than 3m of 'clean' support above ground level?				YES/NO	None
Recreational Area					
5. Is this support, or the associated spans, in the vicinity of a children's playground?				Crosses area/ In close proximity to area/ Not in the immediate area	None
6. Is this support, or the associated spans, in the vicinity of a fishing or sailing location?					
7. Is this support, or the associated spans, in the vicinity of a camping or caravan site?					
8. Is this support, or the associated spans, in the vicinity of common land used for recreational activities?					
9. Is this support, or the associated spans, in the vicinity of an allotment?					
10. Is this support, or the associated spans, in the vicinity of sports ground or stadium?					
11. Is this support, or the associated spans, in the vicinity of a golf course?					
Vocational Area					
12. Is this support, or the associated spans, in the vicinity of a school?				Crosses area/ In close proximity to area/ Not in the immediate area	None
13. Is this support, or the associated spans, in the vicinity of business premises with portable machinery?					
14. Is this support, or the associated spans, in the vicinity of arable land where farm machinery is used?					
Local Environment					
15. Is this support, or the associated spans, in the vicinity of a potential climbing aid?				Crosses area/ In close proximity to area/ Not in the immediate area	None
16. Is this support, or the associated spans, in the vicinity of a 'travellers' site?					
17. Is this support, or the associated spans, in the vicinity of an area prone to vandalism?					
18. Is this support, or the associated spans, in the vicinity of a domestic private garden?					

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Question	Answer	Action
19. Is this support, or the associated spans, in the vicinity of a footpath (inc. Disused railways)?		
Evidence of vandalism or unauthorised access		
20. Is our equipment free from signs of vandalism, unauthorised access or interference?	YES/NO	None
21. Is the area in the immediate vicinity of the line free of evidence of loitering or gatherings of youths? <i>Examples might be drink cans, food wrappings, graffiti, cigarette ends etc. Blown debris should be discounted.</i>	YES/NO	None
22. Is there any equipment or materials on site which may be used to provide unauthorised access to the overhead equipment?	YES/NO	None
Condition of Equipment		
23. Are the clearances to buildings in line with company policy within the site?	YES/NO	Rectify if dangerous
24. Are the clearances to ground in line with company policy within the site?	YES/NO	Rectify if dangerous
25. Are the clearances to trees in line with company policy within the site?	YES/NO	Rectify if dangerous
26. Is the conductor insulation in good condition within the site?	YES/NO	Note or repair as appropriate
27. Is the earthing in good condition within the site?	YES/NO/NOT AT POLE	Note or repair as appropriate
28. Are the supports within the site in good condition?	YES/NO	Note or repair as appropriate
29. Are the stays within the site in good condition?	YES/NO/NOT AT POLE	Note or repair as appropriate
30. Do the stays within the site have stay insulators fitted and located 3m above ground level and below the lowest line on that support?	YES/NO/NOT AT POLE	Note or fit as appropriate
31. Do all the supports within the site have the support number plate present and in good condition?	YES/NO/NOT AT POLE	Note or fit as appropriate
Assessment of Mitigating Measures		
32. Are sufficient danger of death signs fitted to all supports within the site?	YES/NO	Note or fit as appropriate
33. Are anti climbing guards fitted, and in good condition, on all supports within the site?	YES/NO	Note or fit as appropriate
34. Is anti climbing barbed wire fitted and in good condition on all stays within the site?	YES/NO	Note or fit as appropriate
35. Are cable guards fitted and in good condition within the site?	YES/NO	Note or fit as appropriate

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Question	Answer	Action
36. Does the site require inadvertent contact warning signs?	YES/NO	<i>If yes answer questions 37-39</i>
37. Does the site have one or more access points?	YES/NO	<i>Note number</i>
38. Are there warning notices in good condition at each access point? <i>Signs should be fitted where site conditions permit and agreement of landowner has been obtained</i>	YES/NO/NOT AT SITE	<i>Note or fit as appropriate</i>
39. Are there prohibition signs in good condition at appropriate points within the site? <i>Where practicable signs should be fitted 30m either side of the line and 50m apart along the line, where site conditions permit and agreement of landowner has been obtained</i>	YES/NO/NOT AT SITE	<i>Note or fit as appropriate</i>
40. Is the site owner present?	YES/NO	<i>If yes answer questions 41-43</i>
41. Are the site owner's contact details correct?	YES/NO	<i>If no then record</i>
42. Does the site owner have our emergency contact number?	YES/NO	<i>If not then provide</i>
43. Has the site owner refused further advice on safety near electrical lines and equipment?	YES/NO	<i>Note contact details as appropriate</i>

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Appendix E – Application of Standard Mitigating Measures

The table below indicates which standard mitigating measures should be applied to each type of high and medium risk site.

Land Type	Type of Site	Proximity	Level of Risk	Standard Mitigating Measures			
				Additional Safety Signs	Inspection & Risk Assessment	Enhanced ACGs	Inadvertent Contact Signs
Recreational	Children's playgrounds	Both	High	Yes	Yes	Yes	Yes
Recreational	Fishing and Sailing locations (inc. Windsurfing)	Both	High	Yes	Yes	Yes	Yes
Recreational	Camping and Caravan Sites	Both	High	Yes	Yes	Yes	Yes
Recreational	Common land used for recreational activities (e.g. parkland, kite flying, wasteland where youths gather etc)	Crosses	High	Yes	Yes	Yes	Yes
Recreational	Common land used for recreational activities (e.g. parkland, kite flying, wasteland where youths gather etc)	Next to	Medium	Yes	No	No	Yes
Recreational	Allotments	Both	Medium	Yes	No	No	Yes
Recreational	Sports grounds and stadiums	Both	Medium	Yes	No	No	Yes
Recreational	Golf Courses	Both	Medium	Yes	No	No	Yes
Vocational	Schools	Both	High	Yes	Yes	Yes	Yes
Vocational	Business premises with portable machinery	Crosses	High	Yes	Yes	No	Yes
Vocational	Business premises with portable machinery	Next to	Medium	Yes	No	No	No
Vocational	Arable land where farm machinery is used	Both	Medium	Yes	No	No	No
Local	Areas with potential climbing aids	Both	High	Yes	Yes	Yes	No

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				Standard Mitigating Measures			
Land Type	Type of Site	Proximity	Level of Risk	Additional Safety Signs	Inspection & Risk Assessment	Enhanced ACGs	Inadvertent Contact Signs
Local	“Travellers” sites	Both	High	Yes	Yes	Yes	No
Local	Areas prone to vandalism	Both	High	Yes	Yes	Yes	No
Local	Domestic Private Gardens	Crosses	High	Yes	Yes	Yes	No
Local	Domestic Private Gardens	Next to	Medium	Yes	No	No	No
Local	Footpaths (inc Disused railways)	Both	Medium	Yes	No	No	No